

Double Integrals Over Rectangles

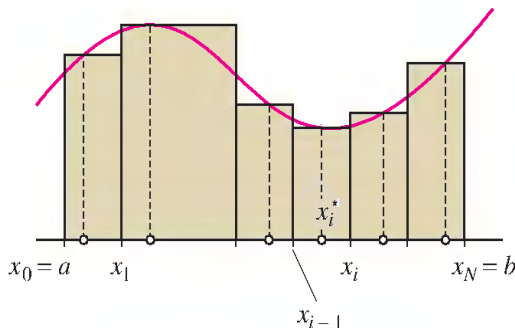
MATH 54
Elementary Analysis II

Recall

If $f(x)$ is defined for $a \leq x \leq b$, we subdivide the interval $[a, b]$ into subintervals of length Δx . Choose x_i^* on each subinterval.

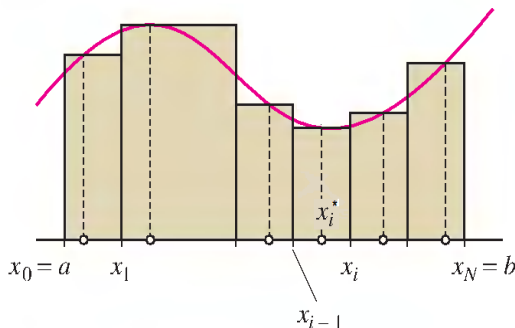
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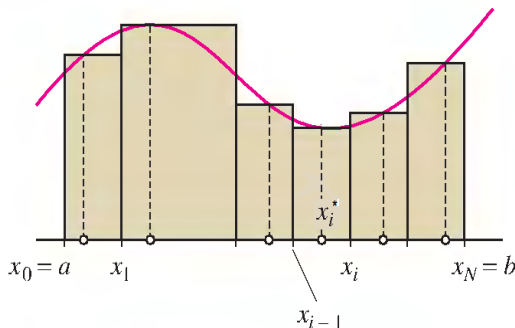
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Volume problem

Let

- $R = [a, b] \times [c, d] = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$, a rectangular region on the xy -plane

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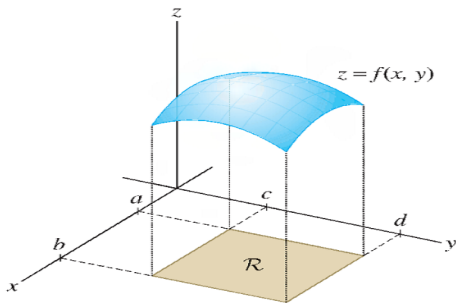
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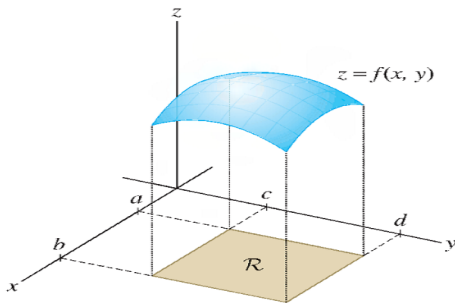


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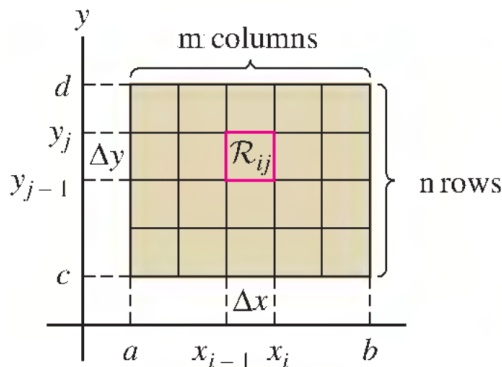
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We want to find: **the volume of the solid under $z = f(x, y)$ over the rectangle R .**



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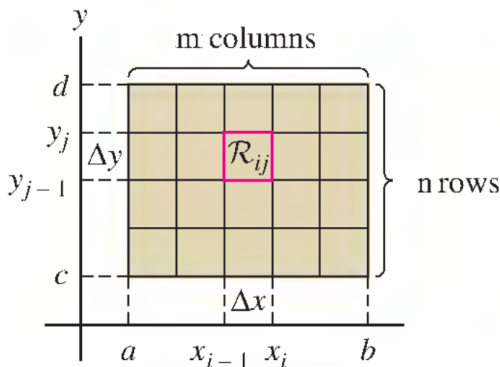
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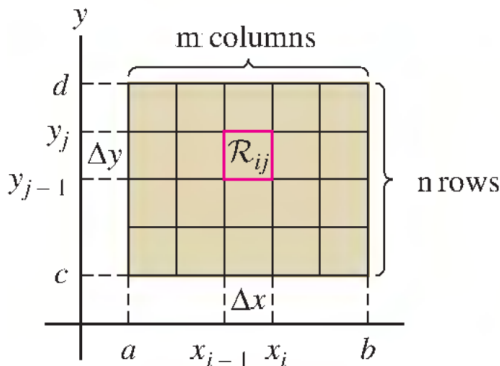


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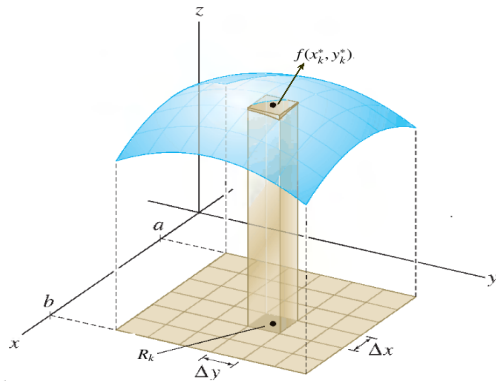
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Let $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ and let $\Delta A_{ij} = \Delta x \Delta y$ be the area of R_{ij} .

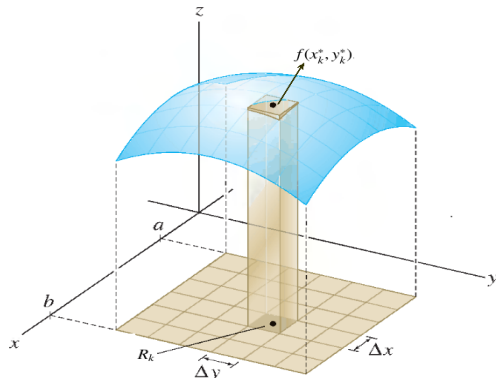


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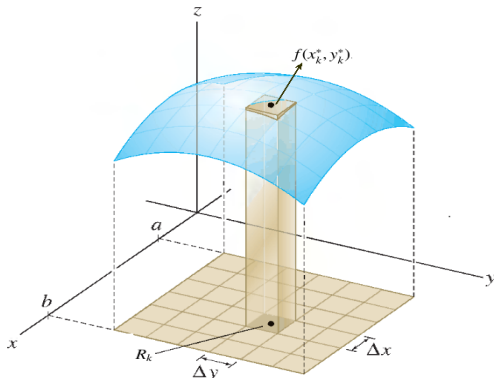
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- Take an arbitrary point (x_k^*, y_k^*) in the k th subrectangle R_k .



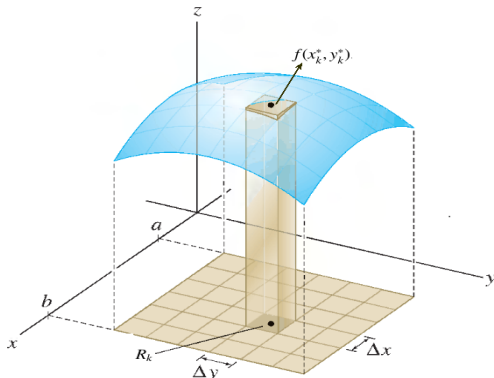
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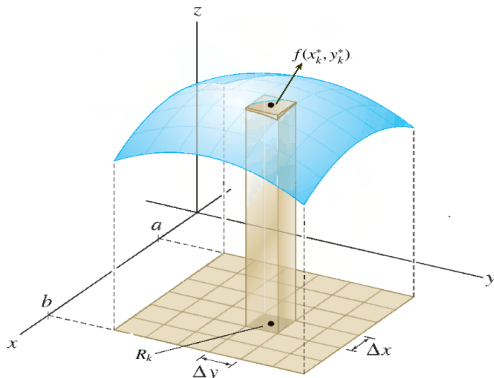
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- The volume of the solid is approximately $\sum_{k=1}^{mn} f(x_k^*, y_k^*) \Delta A_k$.

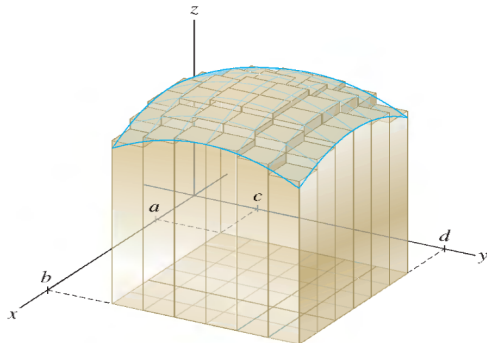


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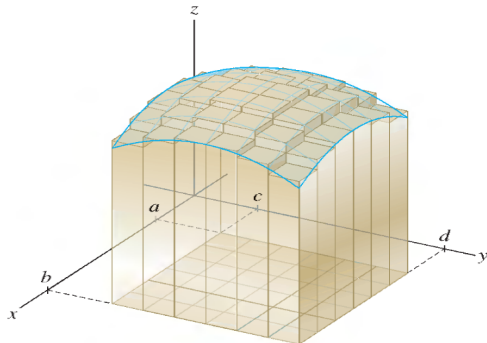
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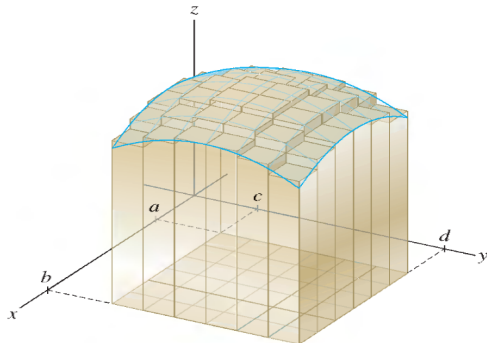
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$$V = \lim_{N \rightarrow \infty} \sum_{k=1}^N f(x_k^*, y_k^*) \Delta A_k.$$

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- The sum

$$S_N = \sum_{k=1}^N f(x_k^*, y_k^*) \Delta A_k$$

may take several values depending on the choice of partitions with subrectangles R_k and the choice of the arbitrary point (x_k^*, y_k^*) in R_k

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$$\iint_R k f(x, y) \, dA = k \iint_R f(x, y) \, dA, \text{ for any real constant } k$$

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where $R = R_1 \cup R_2$ and R_1, R_2 are nonoverlapping.

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Estimate the volume above $R = [0, 2] \times [0, 2]$ and below $z = 16 - x^2 - 2y^2$.

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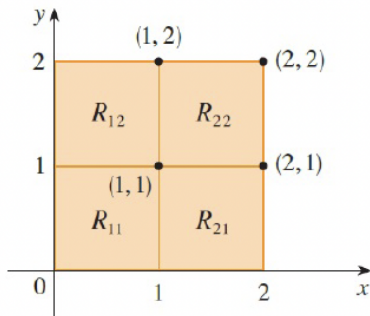
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 $(1, 1), (1, 2), (2, 1), (2, 2)$
- Approximate volume:

$$\begin{aligned} V &\approx \sum f(x_{ij}^*, y_{ij}^*) \Delta A \\ &= f(1, 1) + f(1, 2) + f(2, 1) + f(2, 2) \\ &= 13 + 7 + 10 + 4 = 34 \end{aligned}$$



Therefore, $V \approx 34$ cubic units.

The Midpoint Rule for Double Integrals

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$$\iint_R f(x, y) \, dA \approx \sum_{i=1}^m \sum_{j=1}^n f(\bar{x}_i, \bar{y}_j) \Delta A$$

where $\bar{x}_i$ is the midpoint of $[x_{i-1}, x_i]$ and $\bar{y}_j$ is the midpoint of $[y_{j-1}, y_j]$.

Example of Midpoint Rule

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Use the midpoint rule with $m = n = 2$ to estimate

$$\iint_R (x - 3y^2) dA, \quad R = \{(x, y) \mid 0 \leq x \leq 2, 1 \leq y \leq 2\}.$$

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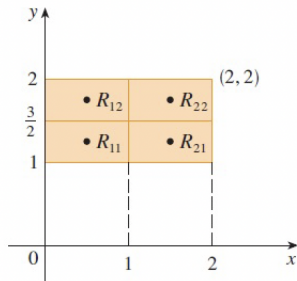
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Midpoints:

$$\left(\frac{1}{2}, \frac{5}{4}\right), \left(\frac{1}{2}, \frac{7}{4}\right), \left(\frac{3}{2}, \frac{5}{4}\right), \left(\frac{3}{2}, \frac{7}{4}\right)$$



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Solution (continued).

Area of each subrectangle:

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$$\begin{aligned}\iint_R (x - 3y^2) dA &\approx \sum_{i=1}^2 \sum_{j=1}^2 f(\bar{x}_i, \bar{y}_j) \Delta A \\&= f(\bar{x}_1, \bar{y}_1) \Delta A + f(\bar{x}_1, \bar{y}_2) \Delta A + f(\bar{x}_2, \bar{y}_1) \Delta A + f(\bar{x}_2, \bar{y}_2) \Delta A \\&= f\left(\frac{1}{2}, \frac{5}{4}\right) \Delta A + f\left(\frac{1}{2}, \frac{7}{4}\right) \Delta A + f\left(\frac{3}{2}, \frac{5}{4}\right) \Delta A + f\left(\frac{3}{2}, \frac{7}{4}\right) \Delta A \\&= \left(-\frac{67}{16}\right) \frac{1}{2} + \left(-\frac{139}{16}\right) \frac{1}{2} + \left(-\frac{51}{16}\right) \frac{1}{2} + \left(-\frac{123}{16}\right) \frac{1}{2} \\&= -\frac{95}{8}\end{aligned}$$

Therefore, $\iint_R (x - 3y^2) dA \approx -\frac{95}{8}$.

Iterated double integrals

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$$\int_a^b I(x) dx = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_a^b \int_c^d f(x, y) dy dx$$

called an **iterated (double) integral**.

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$$I(y) = \int_a^b f(x, y) dx$$

Iterated double integrals

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which is also an iterated double integral.

Iterated double integrals

Example

Evaluate the following iterated integrals.

a. $\int_1^2 \int_0^1 (6x^2y - 1) \, dy \, dx$

b. $\int_0^1 \int_1^2 (6x^2y - 1) \, dx \, dy$

Iterated double integrals

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Evaluate the following iterated integrals.

a. $\int_1^2 \int_0^1 (6x^2y - 1) \, dy \, dx$

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Solution.

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Solution.

$$\begin{aligned} \text{a. } & \int_1^2 \int_0^1 (6x^2y - 1) \, dy \, dx \\ &= \int_1^2 \left(\int_0^1 (6x^2y - 1) \, dy \right) dx \end{aligned}$$

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$$\begin{aligned} \text{a. } & \int_1^2 \int_0^1 (6x^2y - 1) \, dy \, dx \\ &= \int_1^2 \left(\int_0^1 (6x^2y - 1) \, dy \right) dx \\ &= \int_1^2 (3x^2y^2 - y) \Big|_{y=0}^{y=1} dx \end{aligned}$$

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a. $\int_1^2 \int_0^1 (6x^2y - 1) \, dy \, dx$

b. $\int_0^1 \int_1^2 (6x^2y - 1) \, dx \, dy$

$$= \int_1^2 \left(\int_0^1 (6x^2y - 1) \, dy \right) dx$$

$$= \int_1^2 (3x^2y^2 - y) \Big|_{y=0}^{y=1} dx$$

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a. $\int_1^2 \int_0^1 (6x^2y - 1) \, dy \, dx$

$$= \int_1^2 \left(\int_0^1 (6x^2y - 1) \, dy \right) dx$$

$$= \int_1^2 (3x^2y^2 - y) \Big|_{y=0}^{y=1} dx$$

$$= \int_1^2 (3x^2 - 1) dx = (x^3 - x) \Big|_1^2 = 6$$

b. $\int_0^1 \int_1^2 (6x^2y - 1) \, dx \, dy$

$$= \int_0^1 \left(\int_1^2 (6x^2y - 1) \, dx \right) dy$$

$$= \int_0^1 (2x^3y - x) \Big|_{x=1}^{x=2} dy$$

$$= \int_0^1 (14y - 1) dy = (7y^2 - y) \Big|_0^1$$

Iterated double integrals

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Evaluate the following iterated integrals.

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$$= \int_1^2 (3x^2 - 1) dx = (x^3 - x) \Big|_1^2 = 6$$

b. $\int_0^1 \int_1^2 (6x^2y - 1) \, dx \, dy$

$$= \int_0^1 \left(\int_1^2 (6x^2y - 1) \, dx \right) dy$$

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Double integrals over rectangles

Theorem (Fubini's Theorem)

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Example

Evaluate $\iint_R y \sin(xy) \, dA$ where $R: [0, 2] \times [0, \pi]$.

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$$\iint_R y \sin(xy) \, dA = \int_0^\pi \int_0^2 y \sin(xy) \, dx \, dy$$

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$$\begin{aligned} \iint_R y \sin(xy) \, dA &= \int_0^\pi \int_0^2 y \sin(xy) \, dx \, dy \\ &= \int_0^\pi -\cos(xy) \Big|_{x=0}^{x=2} dy \end{aligned}$$

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Double integration and volume

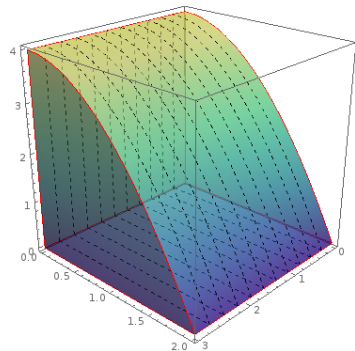
Example

Find the volume of the solid under the surface $f(x, y) = 4 - y^2$ above the region $R: 0 \leq x \leq 3, 0 \leq y \leq 2$ in the xy -plane.

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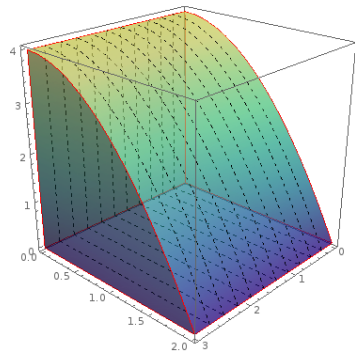
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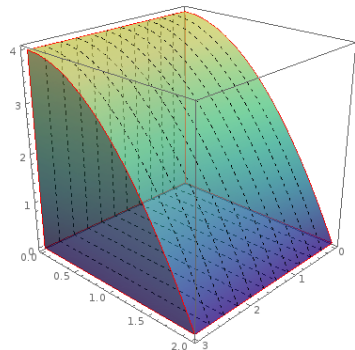
$$V = \int_0^2 \int_0^3 (4 - y^2) dx dy$$



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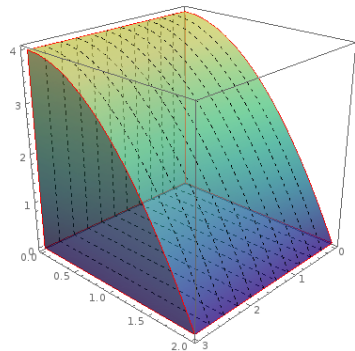
Solution.

$$\begin{aligned} V &= \int_0^2 \int_0^3 (4 - y^2) \, dx \, dy \\ &= \int_0^2 (4x - xy^2) \Big|_{x=0}^{x=3} \, dy \end{aligned}$$

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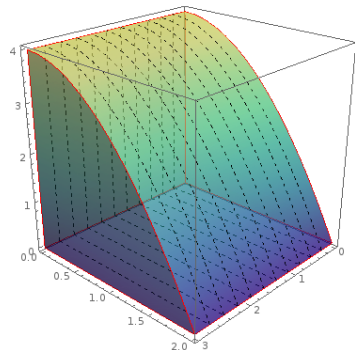
Solution.

$$\begin{aligned} V &= \int_0^2 \int_0^3 (4 - y^2) \, dx \, dy \\ &= \int_0^2 (4x - xy^2) \Big|_{x=0}^{x=3} \, dy \\ &= \int_0^2 (12 - 3y^2) \, dy \end{aligned}$$

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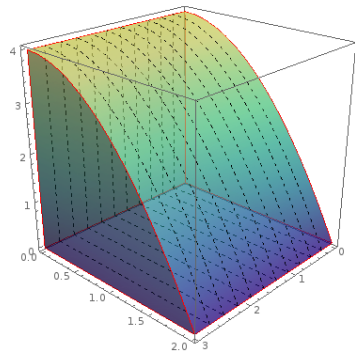
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A neat example using Fubini's theorem

Example

Evaluate the double integral $\iint_R e^x \cos y \, dA$, where $R = [0, 2] \times [0, \frac{\pi}{4}]$.

Solution. Using Fubini's Theorem,

$$\iint_R e^x \cos y \, dA$$

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Evaluate the double integral $\iint_R e^x \cos y \, dA$, where $R = [0, 2] \times [0, \frac{\pi}{4}]$.

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Evaluate the double integral $\iint_R e^x \cos y \, dA$, where $R = [0, 2] \times [0, \frac{\pi}{4}]$.

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$$\iint_R e^x \cos y \, dA = \int_0^2 \int_0^{\pi/4} e^x \cos y \, dy \, dx = \int_0^2 e^x \overbrace{\int_0^{\pi/4} \cos y \, dy}^{\text{constant}} \, dx$$

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Evaluate the double integral $\iint_R e^x \cos y dA$, where $R = [0, 2] \times [0, \frac{\pi}{4}]$.

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Evaluate the double integral $\iint_R e^x \cos y \, dA$, where $R = [0, 2] \times [0, \frac{\pi}{4}]$.

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$$\begin{aligned}\iint_R e^x \cos y \, dA &= \int_0^2 \int_0^{\pi/4} e^x \cos y \, dy \, dx = \int_0^2 e^x \overbrace{\int_0^{\pi/4} \cos y \, dy}^{\text{constant}} \, dx \\&= \int_0^2 e^x \cdot (\text{constant}) \, dx = \text{constant} \cdot \int_0^2 e^x \, dx \\&= \left(\int_0^{\pi/4} \cos y \, dy \right) \left(\int_0^2 e^x \, dx \right) \\&= \left(\int_0^2 e^x \, dx \right) \left(\int_0^{\pi/4} \cos y \, dy \right) \\&= e^x \Big|_0^2 \cdot \sin y \Big|_0^{\pi/4}\end{aligned}$$

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Evaluate the double integral $\iint_R e^x \cos y \, dA$, where $R = [0, 2] \times [0, \frac{\pi}{4}]$.

Solution. Using Fubini's Theorem,

$$\begin{aligned}\iint_R e^x \cos y \, dA &= \int_0^2 \int_0^{\pi/4} e^x \cos y \, dy \, dx = \int_0^2 e^x \overbrace{\int_0^{\pi/4} \cos y \, dy}^{\text{constant}} \, dx \\&= \int_0^2 e^x \cdot (\text{constant}) \, dx = \text{constant} \cdot \int_0^2 e^x \, dx \\&= \left(\int_0^{\pi/4} \cos y \, dy \right) \left(\int_0^2 e^x \, dx \right) \\&= \left(\int_0^2 e^x \, dx \right) \left(\int_0^{\pi/4} \cos y \, dy \right) \\&= e^x \Big|_0^2 \cdot \sin y \Big|_0^{\pi/4} = \frac{\sqrt{2}(e^2 - 1)}{2}.\end{aligned}$$

Factorization via Fubini's Theorem

Suppose $f(x, y) = g(x)h(y)$ and $R = [a, b] \times [c, d]$.

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Result

$$\iint_R g(x)h(y) \, dA = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right)$$

1. Estimate the volume under $z = xy$ over

$$R = \{(x, y) \mid 0 \leq x \leq 6, 0 \leq y \leq 4\}.$$

- a) Use Riemann sum with $m = 3$, $n = 2$, using upper right corners.

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- b) Use midpoint rule with the same partition.

Exercises

1. Evaluate the iterated double integral.

a) $\int_0^1 \int_0^2 xy e^y dx dy$

b) $\int_0^1 \int_{-3}^3 \frac{xy^2}{x^2 + 1} dy dx$

c) $\int_0^{\frac{\pi}{2}} \int_0^1 x \sin(xy) dy dx$

2. Find the volume of the indicated solid

a) under $2x - 3y + z = 6$ and above the rectangle

$$R = \{(x, y) \mid -1 \leq x \leq 0, -2 \leq y \leq 1\}$$

b) under $z = x^2 - y^2$ and above the square $R = [1, 3] \times [-1, 1]$

c) in the first octant bounded by the cylinder $z = 9 - x^2$, the planes $x = 0$, $y = 0$, and $y = 2$

Example

Problem

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Partition points:

$$x = 0, 2, 4, 6 \quad \text{and} \quad y = 0, 2, 4$$

Upper-right sample points:

$$(2, 2), (4, 2), (6, 2), (2, 4), (4, 4), (6, 4)$$

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Evaluate $f(x, y) = xy$:

$$2 \cdot 2 = 4, \quad 4 \cdot 2 = 8, \quad 6 \cdot 2 = 12$$

$$2 \cdot 4 = 8, \quad 4 \cdot 4 = 16, \quad 6 \cdot 4 = 24$$

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$$4 + 8 + 12 + 8 + 16 + 24 = 72$$

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Volume estimate:

$$V \approx 72 \cdot 4 = \boxed{288}$$

Midpoints:

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Evaluate:

$$1 \cdot 1 = 1, \quad 3 \cdot 1 = 3, \quad 5 \cdot 1 = 5$$

$$1 \cdot 3 = 3, \quad 3 \cdot 3 = 9, \quad 5 \cdot 3 = 15$$

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Sum:

$$1 + 3 + 5 + 3 + 9 + 15 = 36$$

Midpoints:

$$x^* = 1, 3, 5 \quad \text{and} \quad y^* = 1, 3$$

Sample points:

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Evaluate:

$$1 \cdot 1 = 1, \quad 3 \cdot 1 = 3, \quad 5 \cdot 1 = 5$$

$$1 \cdot 3 = 3, \quad 3 \cdot 3 = 9, \quad 5 \cdot 3 = 15$$

Sum:

$$1 + 3 + 5 + 3 + 9 + 15 = 36$$

Volume:

$$V \approx 36 \cdot 4 = \boxed{144}$$

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- Upper right Riemann sum: $\boxed{288}$ (overestimate)

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